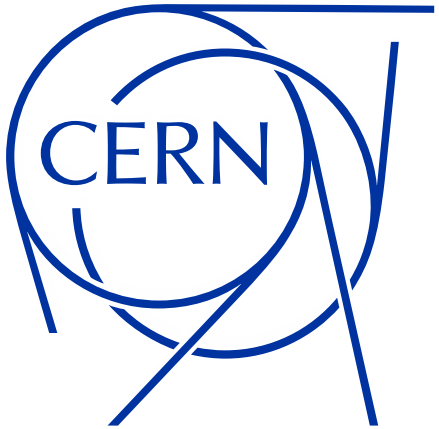


DeMICAF: A Decentralised Medical Imaging Compliance Assessment Framework

INESC TEC

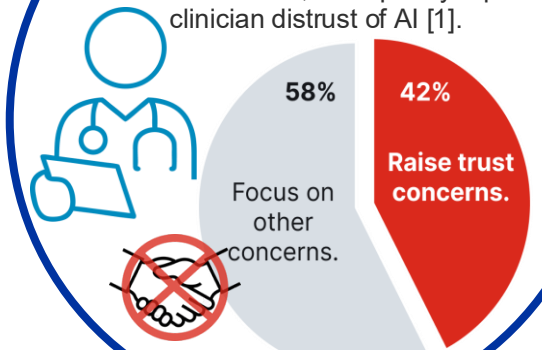
Henrique Romão

1st October 2026

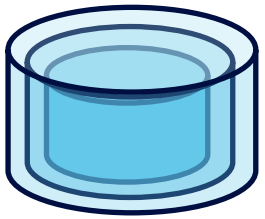
Compliance Assessment

Lack of Clinician's Trust

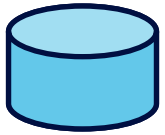
Out of 59 studies, 24 explicitly report clinician distrust of AI [1].



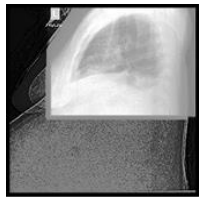
Quantity



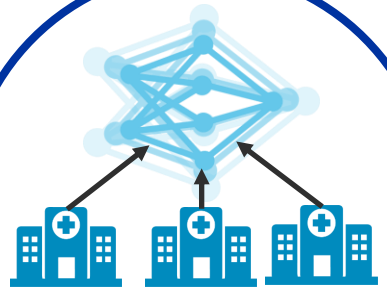
Training Data



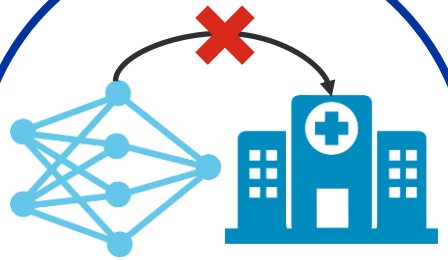
Quality



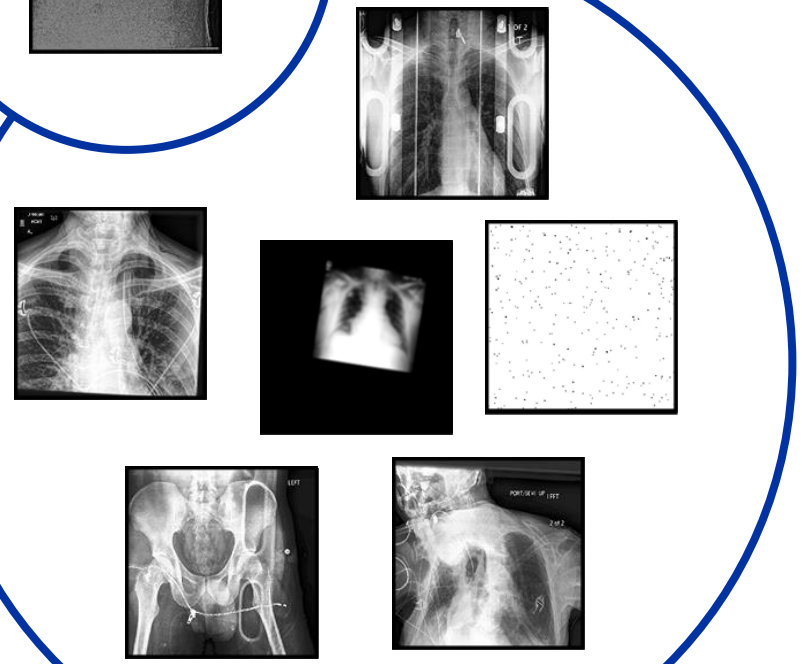
Federated Learning



Limited Clinical Application

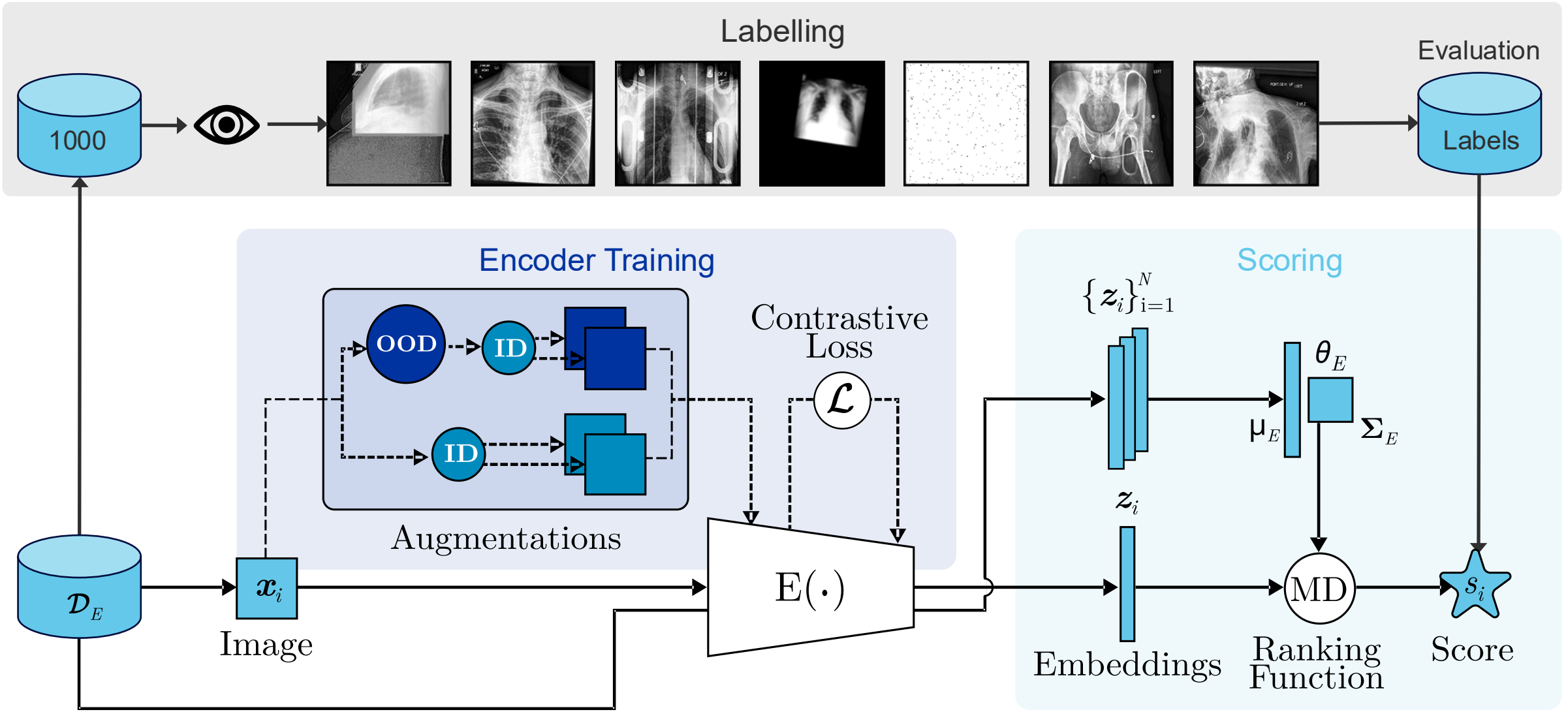


Compliance



[1] Ahmed et al., "A Systematic Review of the Barriers to the Implementation of Artificial Intelligence in Healthcare", Cureus 15(10):e46454, 2023.

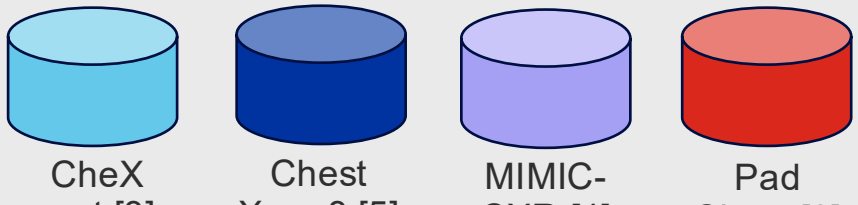
DeepClean [2]

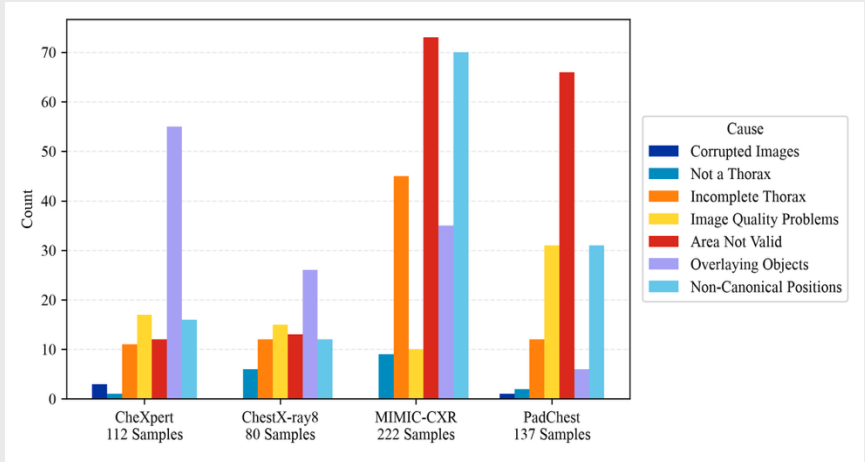


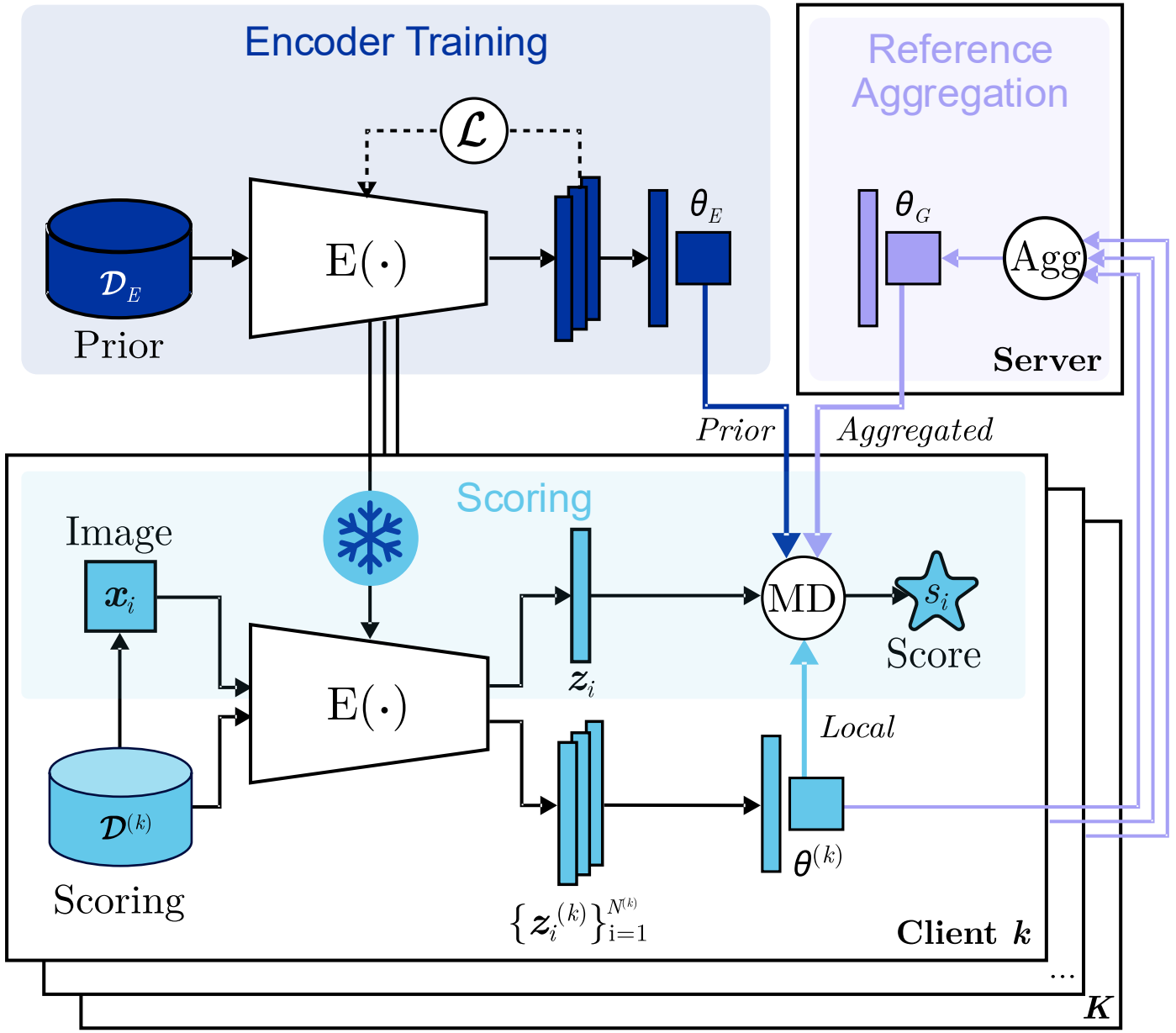
[2] Pereira et al., "DeepClean - Contrastive Learning Towards Quality Assessment in Large-Scale CXR Data Sets," 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 4253-4260, 2024.

DeMICAF

Labelling

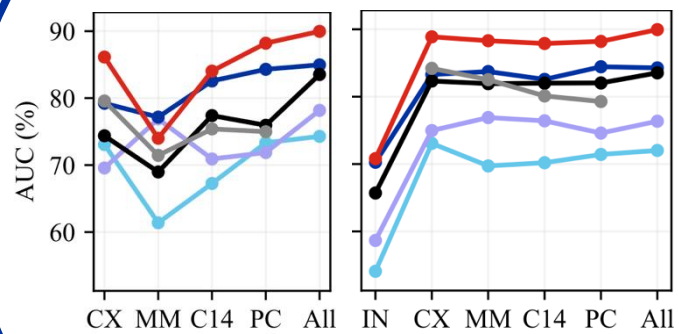






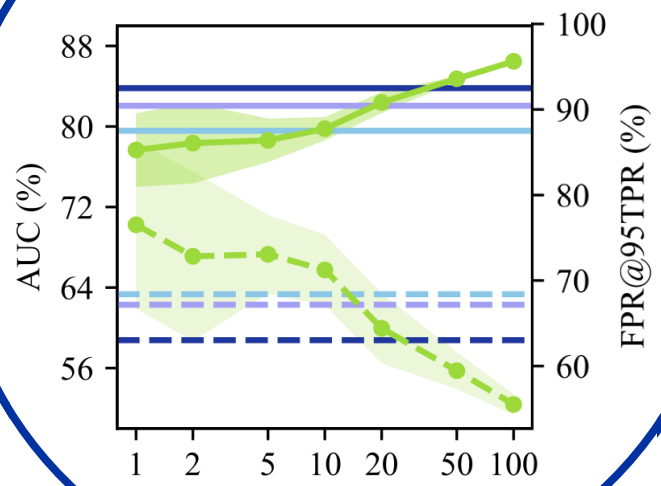
[3] Irvin et al., "CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison," Proceedings of the AAAI Conference on Artificial Intelligence 33(01):590-597, 2019.
[4] Wang et al., "ChestX-ray8: Hospital-Scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases," 2017 IEEE CVPR, pp. 3462-3471, 2017.
[5] Johnson et al., "MIMIC-CXR, a De-identified Publicly Available Database of Chest Radiographs with Free-text Reports," Scientific Data 6(1):317, 2019.
[6] Bustos et al., "PadChest: A Large Chest X-ray Image Dataset with Multi-label Annotated Reports," Medical Image Analysis 66:101797, 2020.

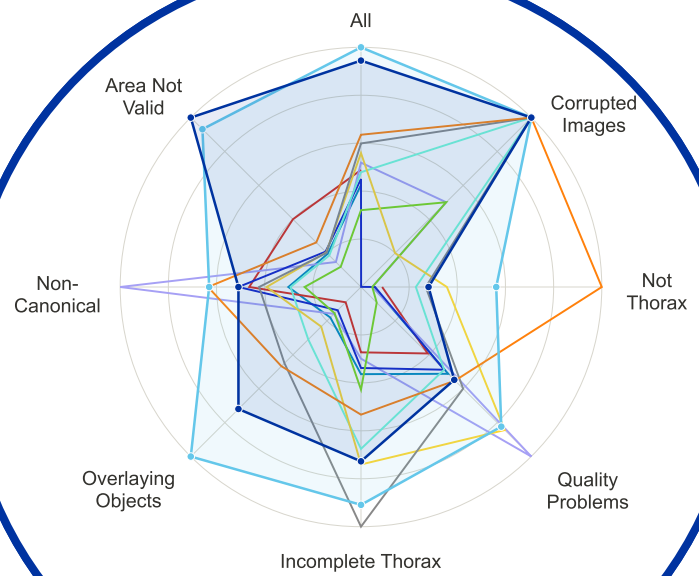
Experiments



Generalization Analysis

Reference Aggregation





Per-Cause Benchmark

Conclusion

- Introduced DeMICAF, decentralising a framework for individual-image compliance assessment.
- Released per-cause compliance annotations for 4,000 chest radiographs from four public datasets.
- Used the annotation for benchmarking both DeMICAF and DeepClean.
- Identified the reference, not the encoder training set, as the dominant design choice.

Future Work

- Extend the framework to other imaging modalities, with their own non-compliance taxonomies.
- Study the differentially private release of client statistics.
- Assess the score's impact on downstream tasks.

